

“INSTITUTO TECNOLÓGICO SUPERIOR RUMIÑAHUI”



**Tecnológico UNIVERSITARIO
“RUMIÑAHUI”**

TEMA: TUTORIAL DE PYSPARK

ASIGNATURA: BIG DATA

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CARRERA: DESARROLLO DE SOFTWARE

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1. Introducción

En el siguiente documento se pretende seguir el tutorial seleccionado para trabajar datos mediante pyspark y usar las funciones especificadas en el mismo

2. Objetivos

Instalar pyspark en Google colab

Configurar el ambiente de trabajo e importar el archivo con el que va a trabajar

Ejecutar las consultas que se especifican en el tutorial

3. Desarrollo

3.1.Instalación

3.1.1. Descargar e instalar el jdk de java en su versión 8



The screenshot shows a Google Colab notebook cell. At the top, there is a toolbar with a pencil icon, a 'Generar' button, a 'Usando...' dropdown menu, and a text input field containing 'a slider using jupyter widgets'. Below the toolbar, a message states: 'Generate está disponible durante un tiempo limitado para los usuarios no suscritos. [Pasarse a Colab Pro](#)'. The code cell contains the following commands:

```
[1] ! apt-get install openjdk-8-jdk-headless -qq > /dev/null
```



```
! java --version
```

Below the code, the output is displayed:

```
openjdk 11.0.21 2023-10-17
OpenJDK Runtime Environment (build 11.0.21+9-post-Ubuntu-0ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.21+9-post-Ubuntu-0ubuntu122.04, mixed mode, sharing)
```

At the bottom of the cell, there is a toolbar with a '+ Código' button and a '+ Texto' button.

3.1.2. Descargar apache spark



The screenshot shows a Google Colab notebook cell. At the top, there is a toolbar with a pencil icon, a 'Generar' button, a 'Usando...' dropdown menu, and a text input field containing 'create a dataframe with 2 columns and 10 rows'. Below the toolbar, a message states: 'Generate está disponible durante un tiempo limitado para los usuarios no suscritos. [Pasarse a Colab Pro](#)'. The code cell contains the following command:

```
! wget -q https://d1cdn.apache.org/spark/spark-3.3.4/spark-3.3.4-bin-hadoop3.tgz
```

Below the code, there is an ellipsis '...' indicating that the output is truncated.

3.1.3. Descomprimir el fichero de apache spark

```
!tar -xf spark-3.3.4-bin-hadoop3.tgz
```

3.1.4. Configurar variables de ambiente

```
[5] import os
```

```
os.environ["JAVA_HOME"] = "/usr/lib/jvm/java-8-openjdk-amd64"  
os.environ["SPARK_HOME"] = "/caontent/spark-3.3.4-bin-hadoop3"
```

3.1.5. Instalar findspark

```
! pip install -q findspark
```

3.1.6. Instalar pyspark

```
! pip install pyspark
```

```
... Collecting pyspark  
  Downloading pyspark-3.5.0.tar.gz (316.9 MB)  
      271.2/316.9 MB 76.6 MB/s eta 0s
```

3.2. Configurar

3.2.1. Crear Spark Session

```
import findspark  
findspark.init()  
from pyspark.sql import SparkSession  
spark = SparkSession.builder.master("local[*]").appName("Ejercicio").getOrCreate()
```

3.2.1.1. Verificamos

```
print(spark)
```

```
<pyspark.sql.session.SparkSession object at 0x79d1243bc190>
```

```
print(spark.version)
```

```
3.3.4
```

3.2.2. Importar los paquetes y clases

```
from pyspark import SparkContext, SparkConf
```

3.3.Cargar el archivo

```
from google.colab import files  
files.upload()
```

```
Services,Web Developer,200000,30,Bangalore,Full Time,Web\r\n3.7,Indegene,Web  
Developer,100000,18,Bangalore,Full Time,Web\r\n4.1,First Student,Web  
Developer,100000,16,Bangalore,Full Time,Web\r\n4.1,Fresher,Web  
Developer,240000,14,Bangalore,Full Time,Web\r\n3.8,Cognizant Technology  
Solutions,Web Developer,300000,13,Bangalore,Full Time,Web\r\n3.3,Infosys,Web  
Developer,200000,12,Bangalore,Full Time,Web\r\n4.1,IBM,Web  
Developer,200000,12,Bangalore,Full Time,Web\r\n3.8,Wipro,Web  
Developer,200000,10,Bangalore,Full Time,Web\r\n3.8,Thomson Reuters,Web  
Developer,300000,9,Bangalore,Full Time,Web\r\n4.1,Accenture,Web  
Developer,400000,8,Bangalore,Full Time,Web\r\n3.8,Amazon,Web  
Developer,100000,8,Bangalore,Full Time,Web\r\n4.0,Lenovo,Web  
Developer,500000,7,Bangalore,Full Time,Web\r\n4.3,Web Developers India,Web  
Developer,300000,6,Bangalore,Full Time,Web\r\n3.4,Happlabs,Web  
Developer,1152000,6,Bangalore,Full Time,Web\r\n3.9,Tatneft,Web  
Developer,200000,6,Bangalore,Full Time,Web\r\n4.6,Giridhara Software  
Services,Web Developer,300000,6,Bangalore,Full Time,Web\r\n4.2,Freelancer,  
Developer,200000,5,Bangalore,Full Time,Web\r\n4.5,Google,Web  
Developer,300000,5,Bangalore,Full Time,Web\r\n4.1,[24]7.ai,Web  
Developer,300000,5,Bangalore,Full Time,Web\r\n3.9,Armentum,Web  
Developer,300000,4,Bangalore,Full Time,Web\r\n3.6,Tech Mahindra,Web  
Developer,300000,4,Bangalore,Full Time,Web\r\n4.4,Innohub Technologies,Web
```

3.3.1. Leer el archivo

Con Show podemos especificar la cantidad de filas que queremos ver

```
df_pyspark = spark.read.csv('Archivo CSV a ser ejecutado.csv', header=True)  
df_pyspark.show(5)
```

```
+-----+-----+-----+-----+-----+-----+-----+  
|Rating|Company Name|Job Title|Salary|Salaries Reported|Location|Employment Status|Job Roles|  
+-----+-----+-----+-----+-----+-----+-----+  
|3.8|Sasken|Android Developer|400000|3|Bangalore|Full Time|Android|  
|4.5|Advanced Millenni...|Android Developer|400000|3|Bangalore|Full Time|Android|  
|4.0|Unacademy|Android Developer|1000000|3|Bangalore|Full Time|Android|  
|3.8|SnapBizz Cloudtech|Android Developer|300000|3|Bangalore|Full Time|Android|  
|4.4|Appoids Tech Solu...|Android Developer|600000|3|Bangalore|Full Time|Android|  
+-----+-----+-----+-----+-----+-----+-----+  
only showing top 5 rows
```

```
df_pyspark = spark.read.csv('Archivo CSV a ser ejecutado.csv', header=True)
df_pyspark.show(22770)
```

4.8	ARK Global Technolo...	Android Developer	700000	2	Bangalore	Full Time	Android
3.7	FuGenX Technologies	Android Developer	100000	2	Bangalore	Full Time	Android
4.6	Outthanking	Android Developer	200000	2	Bangalore	Full Time	Android
2.8	iMars Technologies	Android Developer	150000	2	Bangalore	Full Time	Android
4.7	GreedyGame	Android Developer...	348000	2	Bangalore	Intern	Android
4.1	In Time Tec	Android Developer	600000	2	Bangalore	Full Time	Android
4.3	IndGlobal	Android Developer	300000	2	Bangalore	Full Time	Android
4.1	CYBERNETYX Intera...	Android Developer	864000	2	Bangalore	Full Time	Android
3.7	Karabi Software	Android Developer	120000	2	Bangalore	Full Time	Android
4.3	Keyfalcon Solutions	Android Developer	400000	2	Bangalore	Full Time	Android
3.9	ABCDEF	Android Developer	168000	2	Bangalore	Full Time	Android
4.6	Cumulations Techn...	Android Developer	400000	2	Bangalore	Full Time	Android
4.2	Dcoder	Android Developer...	96000	2	Bangalore	Intern	Android
5.0	powerplay app	Android Developer	1000000	2	Bangalore	Full Time	Android
4.0	Fembuddy	Android Developer	420000	2	Bangalore	Full Time	Android
4.5	Blewsoft Technolo...	Android Developer	240000	2	Bangalore	Full Time	Android
4.2	Across The Globe	Android Developer...	72000	2	Bangalore	Intern	Android
3.9	HARMAN	Android Developer	700000	2	Bangalore	Full Time	Android
3.2	Oust Labs	Android Developer	396000	2	Bangalore	Full Time	Android
4.0	Kaskeid Designs	Android Developer	264000	2	Bangalore	Full Time	Android
3.6	Trukker Technologies	Android Developer	500000	2	Bangalore	Full Time	Android
3.3	Datamatics Global...	Android Developer	1000000	2	Bangalore	Full Time	Android
3.1	Grabhouse	Android Developer	672000	2	Bangalore	Full Time	Android
2.8	Msewa Software So...	Android Developer	276000	2	Bangalore	Full Time	Android
3.8	Honeywell	Android Developer	800000	2	Bangalore	Full Time	Android
3.7	Prematix Software...	Android Developer	200000	2	Bangalore	Full Time	Android
3.4	Provab Technosoft	Android Developer	492000	2	Bangalore	Full Time	Android
3.0	Ai-Connect	Android Developer	600000	1	Bangalore	Full Time	Android
3.3	Idea infinity IT ...	Android Developer	300000	1	Bangalore	Full Time	Android
3.1	Developer On Rent	Android Developer	300000	1	Bangalore	Full Time	Android
4.0	OZOPROP Technologies	Android Developer	432000	1	Bangalore	Full Time	Android
3.7	ThinkTac	Android Developer...	200000	1	Bangalore	Intern	Android
4.3	Ashwin Infottron	Android Developer	1188000	1	Bangalore	Full Time	Android
4.5	Housejoy	Android Developer	500000	1	Bangalore	Full Time	Android
5.0	Proven Solution	Android Developer	500000	1	Bangalore	Full Time	Android
5.0	Blooms Mobility T...	Android Developer	300000	1	Bangalore	Full Time	Android
3.9	DT Consulting	Android Developer	1000000	1	Bangalore	Full Time	Android
4.8	Kartel Technosoft	Android Developer	400000	1	Bangalore	Full Time	Android
3.9	Philips	Android Developer	500000	1	Bangalore	Full Time	Android
3.4	Vensure Employer ...	Android Developer	732000	1	Bangalore	Full Time	Android
4.5	QuHu	Android Developer...	120000	1	Bangalore	Intern	Android
4.1	Software Developers	Android Developer	684000	1	Bangalore	Full Time	Android
4.0	Livwell holding	Android Developer	500000	1	Bangalore	Full Time	Android
3.8	Uolo Technology	Android Developer	800000	1	Bangalore	Full Time	Android
3.3	MSL Learning Systems	Android Developer	400000	1	Bangalore	Full Time	Android
4.8	Fermedicius Labs	Android Developer	200000	1	Bangalore	Full Time	Android
4.1	Dunzo	Android Developer	2800000	1	Bangalore	Full Time	Android
3.7	ABC Tech	Android Developer	852000	1	Bangalore	Full Time	Android
4.3	SpurTree Technolo...	Android Developer	700000	1	Bangalore	Full Time	Android
4.0	Grab (India)	Android Developer	1200000	1	Bangalore	Full Time	Android
3.9	42Gears Mobility ...	Android Developer	900000	1	Bangalore	Full Time	Android
4.1	Tata	Android Developer	400000	1	Bangalore	Full Time	Android
4.1	Kruze	Android Developer	220000	1	Bangalore	Full Time	Android

✓ 4 s completado a las 15:06

Con count podemos saber el numero de registros que tenemos

```
print('Numero de datos en la dataset:', df_pyspark.count())
```

```
Numero de datos en la dataset: 22770
```

Podemos ver que tipo de datos tienen las columnas y debemos modificar los tipos numéricos que aparecen como string

```
df_pyspark.printSchema()
```

```
root
 |-- Rating: string (nullable = true)
 |-- Company Name: string (nullable = true)
 |-- Job Title: string (nullable = true)
 |-- Salary: string (nullable = true)
 |-- Salaries Reported: string (nullable = true)
 |-- Location: string (nullable = true)
 |-- Employment Status: string (nullable = true)
 |-- Job Roles: string (nullable = true)
```

```
3] from pyspark.sql.types import IntegerType, FloatType
df_pyspark = df_pyspark.withColumn("Rating", df_pyspark["Rating"].cast(FloatType()))
df_pyspark = df_pyspark.withColumn("Salary", df_pyspark["Salary"].cast(IntegerType()))
df_pyspark = df_pyspark.withColumn("Salaries Reported", df_pyspark["Salaries Reported"].cast(IntegerType()))

df_pyspark.printSchema()
```

```
root
 |-- Rating: float (nullable = true)
 |-- Company Name: string (nullable = true)
 |-- Job Title: string (nullable = true)
 |-- Salary: integer (nullable = true)
 |-- Salaries Reported: integer (nullable = true)
 |-- Location: string (nullable = true)
 |-- Employment Status: string (nullable = true)
 |-- Job Roles: string (nullable = true)
```

Podemos hacer selects como en sql de una o varias columnas

```
✓ [9] df_pyspark.select('Salary').show(5)
```

```
+-----+
| Salary|
+-----+
| 400000|
| 400000|
|1000000|
| 300000|
| 600000|
+-----+
only showing top 5 rows
```

```
✓ [10] df_pyspark.select('Rating', 'Company Name', 'Job Title', 'Salary').show(5)
```

```
+-----+-----+-----+-----+
|Rating| Company Name| Job Title| Salary|
+-----+-----+-----+-----+
| 3.8|      Sasken|Android Developer| 400000|
| 4.5|Advanced Millenni...|Android Developer| 400000|
| 4.0|      Unacademy|Android Developer|1000000|
| 3.8| SnapBizz Cloudtech|Android Developer| 300000|
| 4.4|Appoids Tech Solu...|Android Developer| 600000|
+-----+-----+-----+-----+
only showing top 5 rows
```

```
✓ [11] df_pyspark.select('Employment Status').distinct().show()
```

```
+-----+
|Employment Status|
+-----+
|      Contractor|
|          Intern|
|        Trainee|
|        Full Time|
+-----+
```

Podemos aplicar condiciones para que se nos muestren los datos y filtrarlos de acuerdo con nuestros requerimientos

```
df_pyspark.filter(df_pyspark['Salary'] > 1000000).show(5)
```

```
+-----+-----+-----+-----+-----+-----+-----+
|Rating| Company Name| Job Title| Salary|Salaries Reported| Location|Employment Status|Job Roles|
+-----+-----+-----+-----+-----+-----+-----+
| 4.3| Ashwin Infotriron|Android Developer|1188000|1|Bangalore| Full Time| Android|
| 4.1|      Dunzo|Android Developer|2800000|1|Bangalore| Full Time| Android|
| 4.0|      Grab (India)|Android Developer|1200000|1|Bangalore| Full Time| Android|
| 4.0|Walmart Global Tech|Android Developer|2300000|1|Bangalore| Full Time| Android|
| 4.0|      Deloitte|Android Developer|1200000|1|Bangalore| Full Time| Android|
+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
```



```
df_pyspark.filter((df_pyspark['Salary'] > 1000000) & (df_pyspark['Employment Status'] == 'Intern')).show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles
4.1	First Student	Android Developer...	2200000		1 Bangalore	Intern	Android
4.3	Josh Talks	Android Developer...	1152000		1 Bangalore	Intern	Android
4.0	Raja Software Labs	Android Developer...	1900000		1 New Delhi	Intern	Android
4.5	Droom	Android Developer...	1500000		1 New Delhi	Intern	Android
3.0	ProtaTech	Android Software ...	1104000		1 New Delhi	Intern	Android

only showing top 5 rows

También con isin podemos filtrar por dos o más tipos en una sola columna

```
[14] df_pyspark.filter(df_pyspark['Company Name'].isin('Unacademy', 'Amazon')).show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles
4.0	Unacademy	Android Developer	1000000		3 Bangalore	Full Time	Android
3.8	Amazon	Android Developer	228000		1 Bangalore	Full Time	Android
3.8	Amazon	Android Developer	300000		1 Bangalore	Full Time	Android
3.8	Amazon	Android Software ...	624000		1 Hyderabad	Full Time	Android
3.8	Amazon	Android Developer...	984000		1 New Delhi	Intern	Android

only showing top 5 rows

```
df_pyspark.filter(df_pyspark['Company Name'].isin('Unacademy', 'Amazon', 'Sasken')).show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles
3.8	Sasken	Android Developer	400000		3 Bangalore	Full Time	Android
4.0	Unacademy	Android Developer	1000000		3 Bangalore	Full Time	Android
3.8	Amazon	Android Developer	228000		1 Bangalore	Full Time	Android
3.8	Amazon	Android Developer	300000		1 Bangalore	Full Time	Android
3.8	Sasken	Android Software ...	600000		1 Bangalore	Full Time	Android

only showing top 5 rows

Podemos agregar y renombrar una nueva columna

```
df_pyspark.filter(df_pyspark['Company Name'].isin('Unacademy', 'Amazon', 'Sasken')).show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles
3.8	Sasken	Android Developer	400000		3 Bangalore	Full Time	Android
4.0	Unacademy	Android Developer	1000000		3 Bangalore	Full Time	Android
3.8	Amazon	Android Developer	220000		1 Bangalore	Full Time	Android
3.8	Amazon	Android Developer	300000		1 Bangalore	Full Time	Android
3.8	Sasken	Android Software ...	600000		1 Bangalore	Full Time	Android

only showing top 5 rows

```
df_pyspark = df_pyspark.withColumn('Rating>=4', df_pyspark['Rating']>=4.0)
df_pyspark.show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles	Rating>=4
3.8	Sasken	Android Developer	400000		3 Bangalore	Full Time	Android	false
4.5	Advanced Millenni...	Android Developer	400000		3 Bangalore	Full Time	Android	true
4.0	Unacademy	Android Developer	1000000		3 Bangalore	Full Time	Android	true
3.8	SnapBizz Cloudtech	Android Developer	300000		3 Bangalore	Full Time	Android	false
4.4	Appoids Tech Solu...	Android Developer	600000		3 Bangalore	Full Time	Android	true

only showing top 5 rows

```
[16] df_pyspark = df_pyspark.withColumnRenamed('Rating>=4', 'RatingGrtThn4')
df_pyspark.show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles	RatingGrtThn4
3.8	Sasken	Android Developer	400000		3 Bangalore	Full Time	Android	false
4.5	Advanced Millenni...	Android Developer	400000		3 Bangalore	Full Time	Android	true
4.0	Unacademy	Android Developer	1000000		3 Bangalore	Full Time	Android	true
3.8	SnapBizz Cloudtech	Android Developer	300000		3 Bangalore	Full Time	Android	false
4.4	Appoids Tech Solu...	Android Developer	600000		3 Bangalore	Full Time	Android	true

only showing top 5 rows

```
[17] df_pyspark = df_pyspark.drop('RatingGrtThn4')
df_pyspark.show(5)
```

Y borrarla

```
df_pyspark = df_pyspark.drop('RatingGrtThn4')
df_pyspark.show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles
3.8	Sasken	Android Developer	400000		3 Bangalore	Full Time	Android
4.5	Advanced Millenni...	Android Developer	400000		3 Bangalore	Full Time	Android
4.0	Unacademy	Android Developer	1000000		3 Bangalore	Full Time	Android
3.8	SnapBizz Cloudtech	Android Developer	300000		3 Bangalore	Full Time	Android
4.4	Appoids Tech Solu...	Android Developer	600000		3 Bangalore	Full Time	Android

only showing top 5 rows

```
[18] df_pyspark.sort('Salary', ascending = False).show(5)
```

Ordenar los valores números

```
df_pyspark.sort(df_pyspark['Salary'].desc(), df_pyspark['Rating'].asc()).show(5)
```

Rating	Company Name	Job Title	Salary	Salaries Reported	Location	Employment Status	Job Roles
3.6	Thapar University	Software Developm...	90000000		1 New Delhi	Full Time	SDE
3.5	Koru UX Design	Senior Front End ...	10000000		1 Pune	Full Time	Frontend
3.6	OASYS Cybernetics	Senior Java Devel...	10000000		1 Chennai	Full Time	Java
3.8	Concentrix	Oracle Database A...	10000000		1 Bangalore	Full Time	Database
3.7	Nityo Infotech	Lead UI Designer, ...	9900000		1 Bangalore	Full Time	Frontend

only showing top 5 rows

Pedir los valores máximos y ordenarlos

```
[20] df_pyspark.groupBy('Location').max('Salary').show(5)
```

```
+-----+-----+
| Location|max(Salary)|
+-----+-----+
| Bangalore| 10000000|
| Kerala| 2900000|
| Madhya Pradesh| 7800000|
| Chennai| 10000000|
| Mumbai| 9800000|
+-----+-----+
only showing top 5 rows
```

```
df_pyspark.groupBy('Location').max('Salary').sort('max(Salary)', ascending = False).show(5)
```

```
+-----+-----+
| Location|max(Salary)|
+-----+-----+
| New Delhi| 90000000|
| Bangalore| 10000000|
| Chennai| 10000000|
| Pune| 10000000|
| Kolkata| 9850000|
+-----+-----+
only showing top 5 rows
```

4. Conclusiones

Podemos usar lenguaje de consultas dentro de nuestra instancia para obtener los valores que queremos dentro de apachespark

Las consultas son muy similares a o que podemos hacer en sql, además podemos modificar los tipos de datos y estructura de las columnas

Fue un poco sencillo familiarizarme con los tipos de consulta que se podían ejecutar en el tutorial

Tuve un pequeño problema en la configuración de la instancia pero se resolvió rápidamente